Fourier-Shell Symbol and Off-Shell Damping for the Swift–Hohenberg Operator

M3 + E57 | Manuscript 1

Self-contained mathematical proof

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Abstract

We show that the Swift–Hohenberg shell operator $\mathcal{A} = (\Delta + k_\star^2)^2$ has the exact Fourier multiplier $p(\xi) = (|\xi|^2 - k_\star^2)^2$. For the dissipative linear equation $\partial_t u = -a\mathcal{A}u$, $a > 0$, its L^2 solution is given by the multiplier $e^{-atp(\xi)}$, and all frequencies separated from the preferred shell $|\xi| = k_\star$ decay at a uniform exponential rate. We prove self-adjointness, an energy identity, whole-space strong decay, and the periodic-domain analogue. We distinguish shell selection from nonlinear pattern selection and localized bound-state existence.

1 Definitions and main theorem

Fix a dimension $d \geq 1$, a preferred wavenumber $k_\star > 0$, and a dissipation coefficient $a > 0$. We use the unitary Fourier transform

$$\widehat{u}(\xi) = \mathcal{F}u(\xi) = (2\pi)^{-d/2} \int_{\mathbb{R}^d} e^{-\mathrm{i}x\cdot\xi} u(x)\, dx, \tag{1}$$

first for $u \in \mathcal{S}(\mathbb{R}^d)$ and then on $L^2(\mathbb{R}^d)$ by Plancherel. Define

$$\mathcal{A} = (\Delta + k_\star^2)^2, \qquad p(\xi) = (|\xi|^2 - k_\star^2)^2. \tag{2}$$

Theorem 1.1 (M3 + E57: exact spectral shell damping)**.** *For every $u_0 \in L^2(\mathbb{R}^d)$, the initial-value problem*

$$\partial_t u = -a(\Delta + k_\star^2)^2 u, \qquad u(0) = u_0, \tag{3}$$

has a unique L^2 mild solution $u \in C([0,\infty); L^2(\mathbb{R}^d))$. The following statements hold:

(i) *The natural L^2 realization of $\mathcal{A}$ is a nonnegative self-adjoint operator with domain $H^4(\mathbb{R}^d)$, and*

$$\boxed{\widehat{\mathcal{A}u}(\xi) = (|\xi|^2 - k_\star^2)^2 \widehat{u}(\xi).} \tag{4}$$

(ii) *The solution satisfies, for almost every $\xi \in \mathbb{R}^d$,*

$$\boxed{\widehat{u}(t,\xi) = e^{-at(|\xi|^2 - k_\star^2)^2} \widehat{u}_0(\xi).} \tag{5}$$

(iii) *For $\delta > 0$, let P_δ be the orthogonal Fourier projection with multiplier $\mathbf{1}_{\Omega_\delta}$, where*

$$\Omega_\delta = \{\xi \in \mathbb{R}^d : \left||\xi|^2 - k_\star^2\right| \geq \delta\}. \tag{6}$$

Then

$$\boxed{\|P_\delta u(t)\|_{L^2} \leq e^{-a\delta^2 t} \|P_\delta u_0\|_{L^2}} \qquad (t \geq 0). \tag{7}$$

(iv) *Every L^2 solution converges strongly to zero:*

$$\lim_{t\to\infty} \|u(t)\|_{L^2} = 0. \tag{8}$$

Nevertheless, for every finite $t \geq 0$, the evolution operator $S(t)u_0 = u(t)$ satisfies

$$\|S(t)\|_{L^2\to L^2} = 1. \tag{9}$$

Thus decay on the whole space has no uniform exponential rate.

2 Fourier symbol, positivity, and the operator domain

Proof of Theorem 1.1(i). For $u \in \mathcal{S}(\mathbb{R}^d)$ and every coordinate j,

$$\widehat{\partial_j u}(\xi) = \mathrm{i}\xi_j \widehat{u}(\xi), \qquad \widehat{\partial_j^2 u}(\xi) = -\xi_j^2 \widehat{u}(\xi). \tag{10}$$

Summing over j gives

$$\widehat{\Delta u}(\xi) = -\,|\xi|^2\, \widehat{u}(\xi). \tag{11}$$

Hence

$$\widehat{(\Delta + k_\star^2)u}(\xi) = (k_\star^2 - |\xi|^2)\widehat{u}(\xi), \tag{12}$$

$$\widehat{(\Delta + k_\star^2)^2 u}(\xi) = (k_\star^2 - |\xi|^2)^2 \widehat{u}(\xi) \tag{13}$$

$$= (|\xi|^2 - k_\star^2)^2 \widehat{u}(\xi), \tag{14}$$

which establishes the asserted Fourier symbol on Schwartz functions.

Under the unitary Fourier transform, the maximal multiplication operator $M_p : f \mapsto pf$ is self-adjoint because p is real-valued, with domain

$$D(M_p) = \{f \in L^2(\mathbb{R}^d) : pf \in L^2(\mathbb{R}^d)\}. \tag{15}$$

Since $p(\xi) \geq 0$, this operator is nonnegative. Moreover,

$$1 + p(\xi)^2 \asymp (1 + |\xi|^2)^4 \qquad \text{for all } \xi \in \mathbb{R}^d, \tag{16}$$

where $f \asymp g$ means that $cg \leq f \leq Cg$ for constants $0 < c \leq C < \infty$ depending only on $k_\star$. To see this, the ratio of the two continuous, strictly positive functions tends to 1 as $|\xi| \to \infty$ and is bounded above and away from zero on any compact set. Therefore

$$\mathcal{F}^{-1} D(M_p) = H^4(\mathbb{R}^d). \tag{17}$$

The operator $\mathcal{A} = \mathcal{F}^{-1} M_p \mathcal{F}$ is thus self-adjoint and nonnegative on H^4; density extends the Fourier identity from $\mathcal{S}$ to this domain.

Finally, for $u \in H^4(\mathbb{R}^d)$, Plancherel yields

$$\langle u, \mathcal{A}u \rangle_{L^2} = \int_{\mathbb{R}^d} p(\xi)\, |\widehat{u}(\xi)|^2 \, d\xi \tag{18}$$

$$= \left\| (\Delta + k_\star^2)u \right\|_{L^2}^2 \geq 0. \tag{19}$$

The associated quadratic form has the larger form domain $H^2(\mathbb{R}^d)$. □

3 The exact dissipative evolution

Proof of Theorem 1.1(ii). For $t \geq 0$ define the Fourier multiplier

$$S(t) = \mathcal{F}^{-1} M_{e^{-atp}} \mathcal{F}. \tag{20}$$

Because $0 < e^{-atp(\xi)} \leq 1$,

$$\|S(t)u_0\|_2 \leq \|u_0\|_2 . \tag{21}$$

The identity $S(t+s) = S(t)S(s)$ follows directly from multiplication of the scalar exponential multipliers. Since $e^{-atp(\xi)} \to 1$ as $t \downarrow 0$, dominated convergence and Plancherel imply

$$\|S(t)u_0 - u_0\|_2 \longrightarrow 0. \tag{22}$$

Thus $(S(t))_{t\geq 0}$ is a strongly continuous contraction semigroup. Its generator is $-a\mathcal{A}$, which gives existence and uniqueness of the mild solution to (3). Explicitly,

$$\widehat{u}(t,\xi) = e^{-atp(\xi)} \widehat{u}_0(\xi) \tag{23}$$

for almost every ξ.

For $t > 0$, the functions $p(\xi)^j e^{-atp(\xi)}$ are bounded for every nonnegative integer j. Accordingly the solution belongs to the operator domain at positive time and satisfies the differential equation strongly there. In particular, high spatial frequencies are smoothed. □

Remark 3.1 (Meaning of a Fourier mode). For general L^2 data, Fourier transforms are equivalence classes defined almost everywhere. The multiplier identity gives an exact decay factor for almost every frequency; a plane wave of exactly fixed frequency is a generalized eigenfunction, not an element of $L^2(\mathbb{R}^d)$.

4 Uniform damping away from the shell

Proof of Theorem 1.1(iii). On Ω_δ as defined in (6),

$$p(\xi) = (|\xi|^2 - k_\star^2)^2 \geq \delta^2. \tag{24}$$

Therefore, for $t \geq 0$,

$$e^{-2atp(\xi)} \leq e^{-2a\delta^2 t} \qquad (\xi \in \Omega_\delta). \tag{25}$$

By Plancherel and (5),

$$\|P_\delta u(t)\|_2^2 = \int_{\Omega_\delta} e^{-2atp(\xi)} \, |\widehat{u}_0(\xi)|^2 \, d\xi \tag{26}$$

$$\leq e^{-2a\delta^2 t} \int_{\Omega_\delta} |\widehat{u}_0(\xi)|^2 \, d\xi \tag{27}$$

$$= e^{-2a\delta^2 t} \, \|P_\delta u_0\|_2^2 . \tag{28}$$

Taking square roots proves (7). □

Corollary 4.1 (Exact preferred shell and near-shell expansion)**.** *The damping exponent*

$$\lambda(\xi) = -a(|\xi|^2 - k_\star^2)^2 \tag{29}$$

is nonpositive everywhere, with $\lambda(\xi) = 0$ exactly when $|\xi| = k_\star$. Writing $|\xi| = k_\star + h$ yields

$$p(\xi) = (2k_\star h + h^2)^2 = 4k_\star^2 h^2 + 4k_\star h^3 + h^4. \tag{30}$$

Thus the leading damping rate near the selected shell is quadratic in the radial deviation h.

5 Whole-space decay, but no uniform exponential gap

Proof of Theorem 1.1(iv). The zero set of the symbol is

$$\Sigma_\star = \{\xi \in \mathbb{R}^d : |\xi| = k_\star\}. \tag{31}$$

For $d = 1$ it consists of two points, and for $d \geq 2$ it is a codimension-one sphere. In each case it has d-dimensional Lebesgue measure zero. Hence $p(\xi) > 0$ for almost every ξ and

$$e^{-2atp(\xi)} \longrightarrow 0 \qquad (t \to \infty) \tag{32}$$

for almost every ξ. As $0 \leq e^{-2atp(\xi)} \, |\widehat{u}_0(\xi)|^2 \leq |\widehat{u}_0(\xi)|^2$ and $\widehat{u}_0 \in L^2$, dominated convergence gives

$$\|S(t)u_0\|_2^2 = \int_{\mathbb{R}^d} e^{-2atp(\xi)} \, |\widehat{u}_0(\xi)|^2 \, d\xi \longrightarrow 0. \tag{33}$$

On the other hand, the norm of a bounded Fourier multiplier on L^2 is the essential supremum of its absolute value, so

$$\|S(t)\|_{L^2 \to L^2} = \operatorname{ess\,sup}_{\xi \in \mathbb{R}^d} e^{-atp(\xi)} = 1. \tag{34}$$

Indeed, every open neighborhood of the nonempty sphere $\Sigma_\star$ has positive Lebesgue measure and contains values of p arbitrarily close to zero. No bound $\|S(t)\|_{L^2 \to L^2} \leq Ce^{-ct}$ with fixed $c > 0$ can therefore hold for all $t \geq 0$. This proves the claimed distinction between strong decay of each L^2 solution and uniform decay of the semigroup. □

Corollary 5.1 (No nonzero exact-shell L^2 state)**.** *If $u \in L^2(\mathbb{R}^d)$ and $\operatorname{supp} \widehat{u} \subseteq \Sigma_\star$ in the essential-support sense, then $u = 0$ almost everywhere.*

Proof. An L^2 Fourier transform supported on a Lebesgue-null set must vanish almost everywhere. Apply Plancherel. □

6 Energy-dissipation identities

Proposition 6.1 (Monotone quadratic energy)**.** *Let $u(t) = S(t)u_0$ with $u_0 \in L^2(\mathbb{R}^d)$. Define, for $t > 0$,*

$$\mathcal{E}[u(t)] = \frac{1}{2} \left\| (\Delta + k_\star^2) u(t) \right\|_2^2. \tag{35}$$

Then, for every $t > 0$,

$$\frac{d}{dt} \mathcal{E}[u(t)] = -a \left\| (\Delta + k_\star^2)^2 u(t) \right\|_2^2 \leq 0, \tag{36}$$

$$\frac{1}{2} \frac{d}{dt} \|u(t)\|_2^2 = -a \left\| (\Delta + k_\star^2) u(t) \right\|_2^2 \leq 0. \tag{37}$$

Proof. Write $L = \Delta + k_\star^2$, so $\mathcal{A} = L^2$. At every positive time the semigroup regularizes sufficiently for the following differentiations and integrations by parts. Since L and $\mathcal{A}$ are self-adjoint and $u_t = -a\mathcal{A}u$,

$$\frac{d}{dt} \frac{1}{2} \|Lu\|_2^2 = \operatorname{Re} \langle Lu, Lu_t \rangle_{L^2} \tag{38}$$

$$= \operatorname{Re} \left\langle L^2 u, u_t \right\rangle_{L^2} \tag{39}$$

$$= -a \|\mathcal{A}u\|_2^2. \tag{40}$$

Similarly,

$$\frac{1}{2}\frac{d}{dt}\|u\|_2^2 = \mathrm{Re}\,\langle u, u_t\rangle_{L^2} \tag{41}$$

$$= -a\,\langle u, \mathcal{A}u\rangle_{L^2} \tag{42}$$

$$= -a\,\|Lu\|_2^2\,. \tag{43}$$

This proves both identities. If $u_0 \in H^4$, they are also valid at $t = 0$; otherwise they hold for $t > 0$ and have the corresponding integrated formulations. □

Remark 6.2 (Gradient-flow convention). Using the *real* Hilbert-space inner product $(u, v)_{\mathbb{R}} = \mathrm{Re}\,\langle u, v\rangle_{L^2}$, the gradient of $\mathcal{E}[u] = \frac{1}{2}\|Lu\|_2^2$ is $\nabla_{\mathbb{R}}\mathcal{E}[u] = \mathcal{A}u$. Thus $u_t = -a\nabla_{\mathbb{R}}\mathcal{E}[u]$ is precisely a negative gradient flow. This avoids possible factors of two from differing conventions for complex Wirtinger derivatives.

7 Periodic-domain counterpart

Proposition 7.1 (Discrete Fourier shells on a flat torus)**.** *Consider a flat torus $\mathbb{T}_L^d = (\mathbb{R}/L\mathbb{Z})^d$. Its Fourier frequencies are $\xi_n = 2\pi n/L$, $n \in \mathbb{Z}^d$. Let Π_0 denote orthogonal projection onto the resonant span*

$$\{e^{\mathrm{i}\xi_n\cdot x} : |\xi_n| = k_\star\}, \tag{44}$$

with $\Pi_0 = 0$ if that span is trivial. Then for a constant $g_L > 0$,

$$\|u(t) - \Pi_0 u_0\|_2 \le e^{-ag_L t}\,\|u_0 - \Pi_0 u_0\|_2\,, \tag{45}$$

where

$$g_L = \min\{(|\xi_n|^2 - k_\star^2)^2 : n \in \mathbb{Z}^d,\ |\xi_n| \neq k_\star\} > 0. \tag{46}$$

In particular, $u(t) \to \Pi_0 u_0$ exponentially in L^2.

Proof. Fourier coefficients solve the decoupled equations

$$\widehat{u}_n(t) = e^{-at(|\xi_n|^2 - k_\star^2)^2}\widehat{u}_n(0). \tag{47}$$

There are finitely many lattice frequencies in each bounded ball, and $(|\xi_n|^2 - k_\star^2)^2 \to \infty$ as $|n| \to \infty$. Consequently, the nonzero values of the discrete symbol have a strictly positive minimum g_L. Parseval's identity gives

$$\|u(t) - \Pi_0 u_0\|_2^2 = \sum_{|\xi_n| \neq k_\star} e^{-2at(|\xi_n|^2 - k_\star^2)^2}\,|\widehat{u}_n(0)|^2 \tag{48}$$

$$\le e^{-2ag_L t}\,\|u_0 - \Pi_0 u_0\|_2^2\,, \tag{49}$$

where the coefficient normalization is chosen so that Parseval holds. Taking square roots proves the claim. □

8 Linear control parameter and limits of the claim

Proposition 8.1 (Linear Swift–Hohenberg threshold)**.** *For the parameterized linear equation*

$$\partial_t u = ru - a(\Delta + k_\star^2)^2 u, \qquad r \in \mathbb{R}, \tag{50}$$

the Fourier growth exponent is

$$\lambda_r(\xi) = r - a(|\xi|^2 - k_\star^2)^2. \tag{51}$$

On the continuum, the spectral bound is r. Thus the linear instability threshold in this centered parameterization is $r_c = 0$; when $r > 0$, the growing frequency band is exactly

$$\left||\xi|^2 - k_\star^2\right| < \sqrt{r/a}. \tag{52}$$

On $\mathbb{T}_L^d$, the first growing mode instead occurs when $r > a \min_{n\in\mathbb{Z}^d}(|\xi_n|^2 - k_\star^2)^2$.

Proof. Applying $\mathcal{F}$ to (50) gives $\partial_t \widehat{u} = \lambda_r(\xi)\widehat{u}$. Since $p \geq 0$ and its minimum is zero on the continuum shell, $\sup_\xi \lambda_r(\xi) = r$; the strict positivity condition $\lambda_r(\xi) > 0$ is equivalent to (52). For a torus, take the maximum over its discrete Fourier frequencies. □

Remark 8.2 (Scope: linear damping versus nonlinear persistence)*.* The exact coefficientwise decay formula (5) concerns the *linear dissipative* evolution with the negative sign in front of the shell operator. Nonlinearities such as $-g\,|u|^2\,u$ couple Fourier frequencies and do not obey independent multiplier decay. The shell symbol alone proves neither nonzero whole-space L^2 bound states nor the existence, uniqueness, or nonlinear stability of localized patterns. In particular, changing the sign to $+a(\Delta + k_\star^2)^2$ reverses the damping and creates high-frequency growth.

Conclusion and classification

The principal identities are

$$\boxed{\begin{aligned} \widehat{(\Delta + k_\star^2)^2 u}(\xi) &= (|\xi|^2 - k_\star^2)^2 \widehat{u}(\xi), \\ \widehat{u}(t,\xi) &= e^{-at(|\xi|^2 - k_\star^2)^2} \widehat{u}_0(\xi), \\ \|P_\delta u(t)\|_2 &\leq e^{-a\delta^2 t}\, \|P_\delta u_0\|_2 . \end{aligned}}$$

Accordingly, **M3 (shell Fourier symbol) is proved**, and **E57 (off-shell spectral damping) is proved** for the stated negative-generator linear flow. Extensions to nonlinear persistence, physical mode confinement, or an independently specified WCT equation require additional hypotheses and independent theorems.